THE RIGHT DERIVED FUNCTOR OF THE INVERSE LIMIT ON N-INDEXED ABELIAN GROUPS HAS THE STANDARD TWO-TERM REPRESENTATION AND VANISHES ABOVE DEGREE ONE

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The counted statement span includes the author's numbered equation labelling the derived functor, delivered verbatim. The author writes 'of R^1 lim_n K^p_n = 0'; and the six references inside the counted proof span name the standard structural facts whose applications stay inside it.

Allowed standard prerequisites (editorial).

- The right-acyclic-subcategory criterion for the existence of a right derived functor. invoked at proof lines 22982-22986. Lets the explicitly built acyclics establish the derived functor and its degree bounds.
- Mittag-Leffler inverse systems and the vanishing of their first derived limit. invoked at proof lines 22983-22984 and 22992. Is the acyclicity statement that the constructed subobject argument delivers.
- The unbounded right derived functor lemma and its proof of the remaining assertions. invoked at proof lines 22994-22999. Extends the derived functor to the unbounded derived category and supplies properties (4) and (5).

Source reference. Original lemma 05T8: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7472–7472. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 07K7: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9647–9647. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 05TA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7474–7474. Cross-chapter navigation pointer into

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Source reference. Original lemma 02N1: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3464–3464. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02MY: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3461–3461. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *The functor* $\lim : Ab(\mathbf{N}) \to Ab$ *has a right derived functor*

$$R\lim : D(Ab(\mathbf{N})) \longrightarrow D(Ab) \tag{0.2.1}$$

As usual we set $R^p \lim(K) = H^p(R\lim(K))$. *Moreover, we have*

1. *for any* (A_n) *in* $Ab(\mathbf{N})$ *we have* $R^p \lim A_n = 0$ *for* $p > 1$,
2. *the object* $R\lim A_n$ *of* $D(Ab)$ *is represented by the complex*
$$\prod A_n \to \prod A_n, \quad (x_n) \mapsto (x_n - f_{n+1}(x_{n+1}))$$
sitting in degrees 0 *and* 1,
3. *if* (A_n) *is ML, then* $R^1 \lim A_n = 0$, *i.e.,* (A_n) *is right acyclic for* lim,
4. *every* $K^\bullet \in D(Ab(\mathbf{N}))$ *is quasi-isomorphic to a complex whose terms are right acyclic for* lim, *and*
5. *if each* $K^p = (K^p_n)$ *is right acyclic for* lim, *i.e., of* $R^1 \lim_n K^p_n = 0$, *then* $R\lim K$ *is represented by the complex whose term in degree* p *is* $\lim_n K^p_n$.

Proof. Let (A_n) be an arbitrary inverse system. Let (B_n) be the inverse system with

$$B_n = A_n \oplus A_{n-1} \oplus \ldots \oplus A_1$$

and transition maps given by projections. Let $A_n \to B_n$ be given by $(1, f_n, f_{n-1} \circ f_n, \ldots, f_2 \circ \ldots \circ f_n)$ where $f_i : A_i \to A_{i-1}$ are the transition maps. In this way we see that every inverse system is a subobject of a ML system (Homology, Section 02MY). It follows from Derived Categories, Lemma 05T8 using Homology, Lemma 02N1 that every ML system is right acyclic for lim, i.e., (3) holds. This already implies that RF is defined on $D^+(Ab(\mathbf{N}))$, see Derived Categories, Proposition 05TA. Set $C_n = A_{n-1} \oplus \ldots \oplus A_1$ for $n > 1$ and $C_1 = 0$ with transition maps given by projections as well. Then there is a short exact sequence of inverse systems $0 \to (A_n) \to (B_n) \to (C_n) \to 0$ where $B_n \to C_n$ is given by $(x_i) \mapsto (x_i - f_{i+1}(x_{i+1}))$. Since (C_n) is ML as well, we conclude that (2) holds (by proposition reference above) which also implies (1). Finally, this implies by Derived Categories, Lemma 07K7 that $R\lim$ is in fact defined on all of $D(Ab(\mathbf{N}))$. In fact, the proof of Derived Categories, Lemma 07K7 proceeds by proving assertions (4) and (5). □